



Sustainable Signalized Intersection Management Model in Border Areas

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Abstract: Signalized intersections often cause congestion. Factors contributing to congestion include the high proportion of intersections operating beyond their capacity and malfunctioning traffic signals. This study aims to develop a sustainable model for signalized intersections in border areas. Primary data were collected by calculating the number of motorized vehicles at each intersection and obtaining expert opinions through focus group discussions (FGDs) to determine the relevant attributes and dimensions. The results from the five signalized intersections show that the Salabenda intersection achieved the highest technological dimension score (72.33%), indicating that technological sustainability is well developed and measurable. The Semplak intersection also demonstrated a strong technological dimension (62.36%), reflecting the implementation of measurable traffic management technology. The Bubulak intersection obtained a social dimension score of 57.71%, indicating that social sustainability, including accessibility and public service aspects, is relatively well implemented. The POMAD intersection achieved an ecological dimension score of 59.44%, showing that environmental considerations are becoming more prominent in traffic management. In the institutional dimension, the Bubulak intersection scored 50.00%, indicating that institutional coordination and management are moderately measurable. Meanwhile, the Ciawi intersection obtained the lowest score in the economic dimension (42.97%), suggesting that economic sustainability still requires improvement. The new simulation model produced four scenarios: sustaining intersection functions through technology, collaborative management of transportation infrastructure for both road and rail systems, and sustainable accessibility control based on the characteristics of border areas. Strengthening the institutional dimension, including interregional cooperation, requires more effective policy and decision-making processes. The novelty of this research lies in the system model node (SYSMODE) concept. This single-point system concept provides benefits across all five sustainability dimensions. The implementation of this five-dimensional model at each intersection must be carried out properly and in a controlled manner. The proposed model is expected to improve the performance of signalized intersections for both road and rail transportation, supported by complete and measurable infrastructure facilities in border areas.

Keywords: Signalized intersections; Focus group discussions; Attributes; Dimensions; System model node

1 Introduction

The movement of people and goods is characterized by the interconnected flow of mobility through transportation infrastructure and facilities within an integrated transportation system. The implementation of such a transportation system is supported by four interrelated concepts that contribute to an integrated framework: multimodal, multidimensional, multistakeholder, and multi-agency approaches. These multi-characteristics illustrate a directed interconnection that supports effective and well-functioning intersection management. Important factors in transportation at intersections include traffic signs, vehicles, and drivers, while roads serve as the primary supporting infrastructure for traffic operations [1, 2].

Vehicles, roads, and drivers are the most important components of transportation systems. Vehicles, roads, and humans as controllers enable movement from one location to another in a faster and more efficient manner. This efficiency in movement requires improved vehicle performance and transportation infrastructure [3, 4]. Efforts to minimize new infrastructure development and optimize existing infrastructure are combined with initiatives to create attractive urban core areas by controlling transportation demand. This approach involves traffic engineering in city centers through comprehensive traffic management strategies aimed at improving the performance of the transportation sector [5, 6].

Bogor, as a buffer zone for Jakarta, is expected to maintain its unique attractiveness both now and in the future, thereby supporting continuous urban growth in the Bogor area. In addition to the Bogor Botanical Gardens, one of the world's renowned urban forests, Bogor is also part of the Greater Jakarta (Jakarta, Bogor, Depok, Tangerang, and Bekasi) metropolitan area. Therefore, traffic management in Bogor City and Bogor Regency must be directed toward maintaining urban mobility and preventing uncontrolled transportation congestion. The objective of this research is to formulate a sustainable management model for signalized intersections in border areas. The study utilizes both primary and secondary data. The novelty of this research lies in the development of a simulation model with four scenarios using the performance of the system model node (SYSMODE) to support the improvement of intersection functionality in the border area between Bogor City and Bogor Regency.

1.1 Intersection Management Strategy

Strategies implemented simultaneously in intersection management to create reliable traffic engineering include inter-regional coordination, synchronization of traffic signal timing, regulation of vehicle movement and traffic volume, prioritization of public transportation and emergency vehicles (such as ambulances), integration of road networks in border areas, improvement of accessibility and driving safety, and regular monitoring of intersections [7, 8].

1.2 Inter-Regional Coordination

Regional development is an effort to strengthen the interdependence between economic, social, environmental, and natural resource systems. Regional areas are defined using a systemic functional concept that links functions and components within an area according to their functional roles [9, 10]. In addition, restructuring urban spatial plans for specific activities often requires land conversion. This process is influenced by the economic value of land use, particularly the net profit generated per unit area within a certain period. Land conversion aims to optimize the distribution and utilization of regional resources [11–13]. Furthermore, the development of an Origin–Destination matrix (ODM) for transportation analysis should be based on transportation problem-solving techniques. Data collection is conducted through questionnaires and travel destination surveys of public transportation users during their trips [14, 15].

2 Research Methods

2.1 Research Location

The research location is in Bogor City and Bogor Regency. Bogor Regency was selected as a buffer district for Bogor City and serves as an access point for motorized vehicles from outside Bogor City, as it serves as a buffer between the city's borders. Five intersections are located: the Salabenda signalized intersection in Kemang, the Semplak signalized intersection in Semplak, the Bubulak signalized intersection in Bubulak, the POMAD signalized intersection in Ciluer, and the Ciawi signalized intersection in Ciawi. Next, a road network map is displayed according to Figure 1.

2.2 Data Sources

This study used both primary and secondary data sources. Primary data were collected through field surveys, observations at the research locations, and interviews with experts. The experts included academics, researchers, and representatives from the Indonesian Transportation Society (ITS). Focus group discussions (FGDs) involving government representatives were also conducted to support the primary data collection. The research locations in Bogor Regency included the Salabenda and Ciawi signalized intersections, while the locations in Bogor City included the Semplak, Bubulak, and POMAD signalized intersections. Secondary data were obtained from journals, proceedings, reports, official documents, and transportation regulations related to Bogor City and Bogor Regency. The selected signalized intersections are presented in Figure 2.

2.3 Data Collection Method

Primary data were collected by identifying existing infrastructure within the research area. After determining the study locations, the research focus was established. The collected data included land area, land use, ownership status, and supporting facilities that could be used as policy recommendations [16]. Field measurements were conducted

using a GPS, cameras, and measuring tapes over six days. Area mapping was carried out using a Phantom 3 Pro drone for six days. A theodolite was also used to verify measurement accuracy at the study locations. In addition, vehicle speed, traffic volume, and density were measured at the five signalized intersections to support the proposed development recommendations [17].

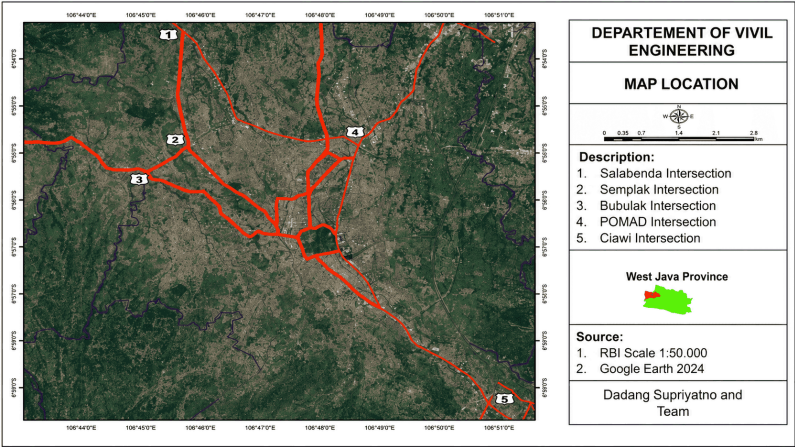


Figure 1. Research location

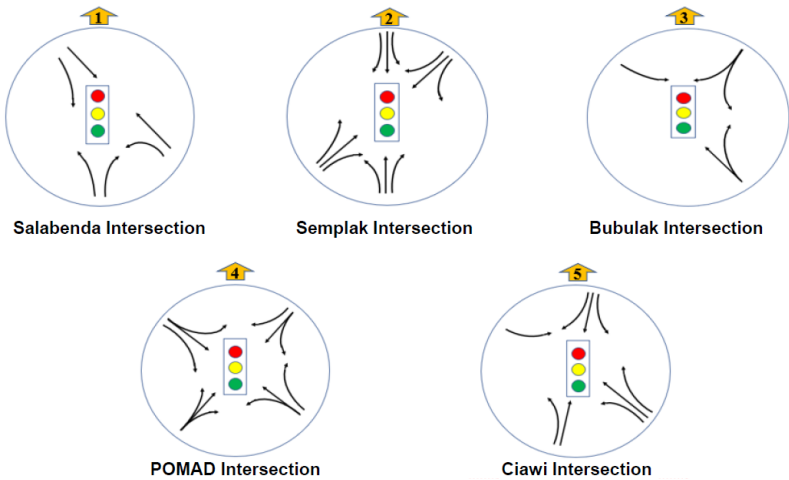


Figure 2. Research locations at five signalized intersections

2.4 Data Analysis Method

Data analysis was conducted using an expert system approach involving five experts. The selection criteria for the experts were as follows:

- Two experts from academic and research backgrounds.
- Two experts with experience as policymakers or decision-makers.
- One transportation specialist from the ITS.

The number of experts was determined based on the recommended range of 3–7 respondents in expert-based assessments [18].

2.5 Data Types and Sources

This study used primary data obtained from experts through FGDs. Five experts participated in the discussion over two days to determine the dimensions, criteria, and attributes used in the sustainability assessment. Various opinions emerged during the discussion; however, a consensus was reached based on field observations. The experts agreed that the sustainability assessment should identify key attributes requiring policy intervention. The assessment was conducted using five dimensions: technological, economic, ecological, social, and institutional.

The sustainability analysis applied the Multi-Dimensional Scaling (MDS)-Rapfish (Rapid Appraisal for Fisheries) method using Multi-Criteria Analysis (MCA). Rapfish uses an MDS ordination technique to measure sustainability

levels based on selected attributes. The dimensions and attributes were determined through expert agreement and literature review.

2.5.1 Technological dimension

The attributes in the technological dimension were determined through literature reviews, transportation guidelines, regulations, and consultations with policymakers. Each attribute was selected to represent activities supporting technological sustainability at signalized intersections. The analysis used 11 attributes, 5 classifications, and a scoring scale of 0–3.

2.5.2 Economic dimension

The economic dimension attributes were identified through FGDs with experts and policymakers, followed by a literature review related to intersection performance. Each attribute reflected activities supporting economic sustainability. The analysis used 8 attributes, 6 classifications, and a scoring scale of 0–3.

2.5.3 Ecological dimension

The ecological dimension attributes were determined through FGDs and literature reviews related to environmental conditions at intersections. Each attribute represented activities supporting ecological sustainability. The analysis used 7 attributes, 5 classifications, and a scoring scale of 0–3.

2.5.4 Social dimension

The social dimension attributes were identified through FGDs with experts and policymakers, supported by a literature review on road user behavior. Each attribute described activities supporting social sustainability. The analysis used 7 attributes, 7 classifications, and a scoring scale of 0–3.

2.5.5 Institutional dimension

The institutional dimension attributes were determined through FGDs and literature reviews related to institutional and management aspects. Each attribute represented activities supporting institutional sustainability. The analysis used 9 attributes, 5 classifications, and a scoring scale of 0–3.

Table 1 presents selected attributes, indicators, and assessment scales used in the MDS-Rapfish analysis at signalized intersections in border areas.

However, Table 1 only displays a portion, as displaying the entire table would take up many pages. The explanation of the dimensions is clear and comprehensive.

Next, the stages of expert assessment, how to make decisions, and how to give scores are shown in Table 2.

2.6 Framework

The following illustrates the framework for sustainable signalized intersection management in border areas, as shown in Figure 3.

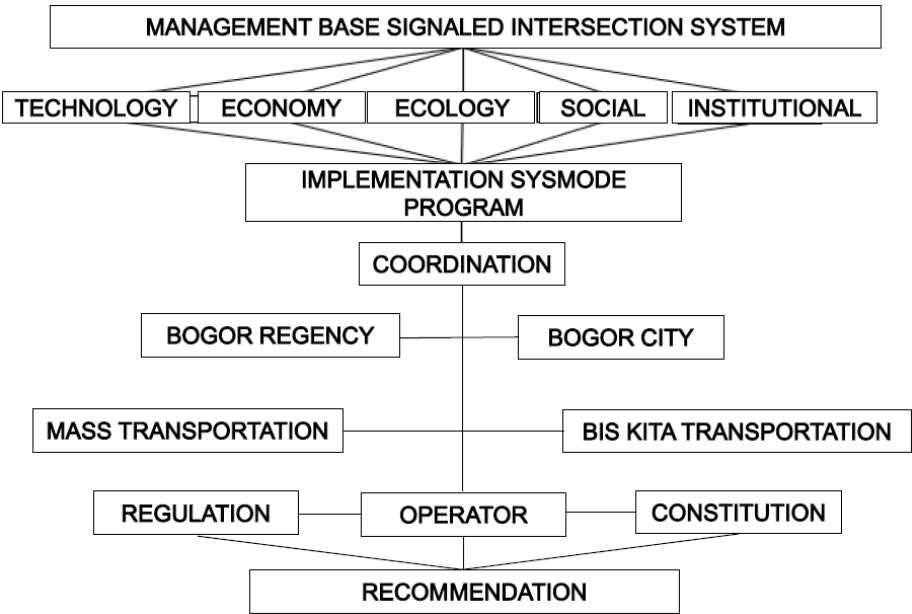


Figure 3. Framework for a sustainable signalized intersection management model in border areas

Table 1. Multi-Dimensional Scaling (MDS)-Rapfish method using multi-criteria analysis (MCA)

Dimension	Field Attributes	Assessment Indicators	Assessment Scale
Technology	Traffic light condition	Signal function and cycle time	0 = Damaged, 1 = Poor, 2 = Good, 3 = Very good
	Vehicle queue length	Queue conditions during peak hours	0 = Very long, 1 = Long, 2 = Medium, 3 = Short
	Delay level	Vehicle delay time at intersection	0 = Very High, 1 = High, 2 = Medium, 3 = Low
Economics	Vehicle operating costs	Effect of delay on fuel consumption	0 = Very high, 1 = High, 2 = Medium, 3 = Low
	Nearby trading activity	Impact of intersection on economic access	0 = Disrupted, 1 = Poor, 2 = Fairly smooth, 3 = Smooth
	Efficiency of goods distribution	Smoothness of logistics transportation	0 = Very disrupted, 1 = Disrupted, 2 = Fair, 3 = Good
Ecology	Vehicle emissions	Queue pollution level	0 = Very high, 1 = High, 2 = Moderate, 3 = Low
	Traffic noise	Vehicle noise intensity	0 = Very noisy, 1 = Noisy, 2 = Moderate, 3 = Low
	Intersection drainage	Intersection flooding conditions	0 = Frequently flooded, 1 = Occasionally flooded, 2 = Smooth, 3 = Very good
Social	Road user safety	Conflict and accident level	0 = Very high, 1 = High, 2 = Low, 3 = Very low
	Pedestrian comfort	Sidewalk and crossing facilities	0 = Not available, 1 = Poor, 2 = Good, 3 = Very good
	Road user compliance	Traffic Violation Level	0 = Very high, 1 = High, 2 = Moderate, 3 = Low
Institutional	Agency supervision	Field monitoring intensity	0 = none, 1 = Infrequent, 2 = Routine, 3 = Very routine
	Inter-agency coordination	Intersection management cooperation	0 = None, 1 = Insufficient, 2 = Sufficient, 3 = Good
	Policy support	Availability of regulations and programs	0 = None, 1 = Limited, 2 = Sufficient, 3 = Complete

Table 2. The stages of expert assessment, the decision-making process, and scoring methods

Stage	Implementation
Field observations	Direct observation of signalized intersection conditions.
Technical measurements	Surveys of delays, queues, and traffic volumes.
Expert interviews	Assessments by academics, the transportation agency, and practitioners.
Focus group discussions (FGDs)	Perception alignment of attribute scores.
Score summary	Final scores are obtained from the average of expert assessments.
Rapfish analysis	All scores are entered into the MDS Rapfish analysis.

As seen in Figure 3, the intervention pattern through regulations from the central and regional governments, their implementation in policies in the field to overcome congestion at each intersection, the expected output of coordination between regions is more often carried out through agglomeration, and the application of SYSMODE.

2.7 Modeling Sustainable Signalized Intersection Management in Border Areas

To determine the sustainability level/status at five research locations, the MDS analysis approach was used, using Rapfish software with the Non-Parametric Multidimensional Scaling approach [19–21].

In urban transportation planning, Rapfish MDS supports strategic decision-making by identifying priority dimensions requiring policy intervention. The method assists planners in evaluating infrastructure performance, accessibility, traffic management, environmental impacts, governance effectiveness, and economic feasibility simultaneously [22, 23]. The sustainability index facilitates comparison among transportation corridors, intersections, or cities and provides a scientific basis for developing sustainable transportation policies. Because Rapfish MDS is relatively fast, flexible, and capable of integrating qualitative expert assessments with quantitative

indicators, it has become an effective decision-support tool for transportation sustainability assessments and long-term urban mobility planning [24, 25].

To compile an index of the sustainability status of the traffic engineering and transportation system model in the study area for each dimension and its attributes, an assessment score for each dimension is obtained, with a scale ranging from 0% to 100%, from worst (bad) to best (good). The relationship between the assessment index categories is shown in Table 3.

Table 3. Assessment index categories

No.	Index Value	Sustainability Categories
1	0–25	Poor: unsustainable
2	26–50	Poor: less sustainable
3	51–75	Fair: quite sustainable
4	76–100	Good: very sustainable

To determine key factors for the traffic engineering and transportation system model in the study area, a prospective analysis approach was used, analyzing the problem using an expert system and restructuring the decision-making process using a different approach.

2.8 Primary Data Included

Structured interviews with experts, including two academics, two from relevant institutions/agencies, and one from the ITS.

Secondary data included:

- Accessibility of vehicle lanes around intersections,
- Side obstacles that impede vehicle movement,
- Function and suitability of traffic signs at intersections,
- Readiness of each region to manage the sustainability of traffic signs at intersections.

3 Results and Discussion

3.1 Results and Discussion of the Technology Dimension

This paper explains the technology dimension in sustainable traffic engineering in transportation systems using multidimensional scaling in a lower dimension while maintaining the distance properties between the studied sections. The technology dimension describes 11 attributes: side barriers, intersection capacity, road width, pedestrian width, traffic signs, green time, saturation level, queue length, bus stop/stop distance, service level, and ATCS (Area Traffic Control System) monitoring in intersection supervision. The results of the sustainability analysis of the technology dimension are, respectively, the Salabenda signalized intersection (72.33%), the Semplak signalized intersection (62.36%), the Bubulak signalized intersection (50.99%), the POMAD signalized intersection (55.70%), and the Ciawi signalized intersection (64.15%). These results indicate that the Salabenda signalized intersection performs best in the technology dimension, with the highest sustainability score among the five intersections, as shown in Figure 4.

The sustainability index, using ordination, also generates leverage to detect dominant attributes. Leverage indicates the change in ordination due to the removal of one attribute for one sensitivity, with values ranging from 2% to 6%, measured by the root mean square. The results are shown in Figure 5.

The degree of saturation significantly influences the technology dimension, at 5.35%. The technology dimension leverages the level of service.

3.2 Results and Discussion of the Economic Dimension

The economic dimension describes eight attributes: contribution to border public transportation fees, management and maintenance of traffic signs, travel costs, distance to the city center, travel time to the city center, availability of regional budget funds, number of underprivileged residents, and the role of the private sector/region-owned enterprises in transportation management. The results of the economic-dimension sustainability analysis were as follows: the Salabenda signalized intersection (42.97%), the Semplak signalized intersection (39.67%), the Bubulak signalized intersection (55.89%), the POMAD signalized intersection (48.29%), and the Ciawi signalized intersection (62.25%). These results indicate that the Ciawi signalized intersection performs best in the economic dimension, with the highest sustainability score among the five intersections, as shown in Figure 6.

Furthermore, the economic dimension of leverage is presented in Figure 7. The economic dimension leverages waiting time and distance to the city center.

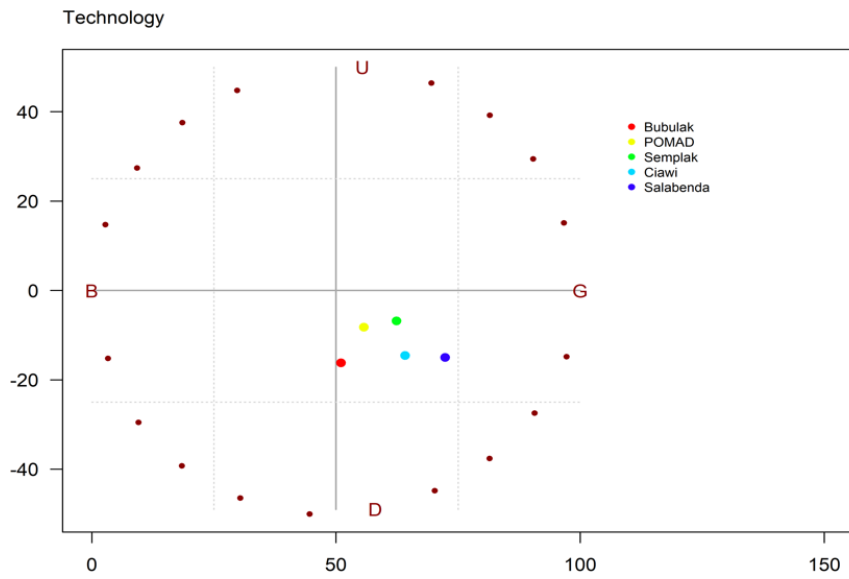


Figure 4. Technology sustainability ordination

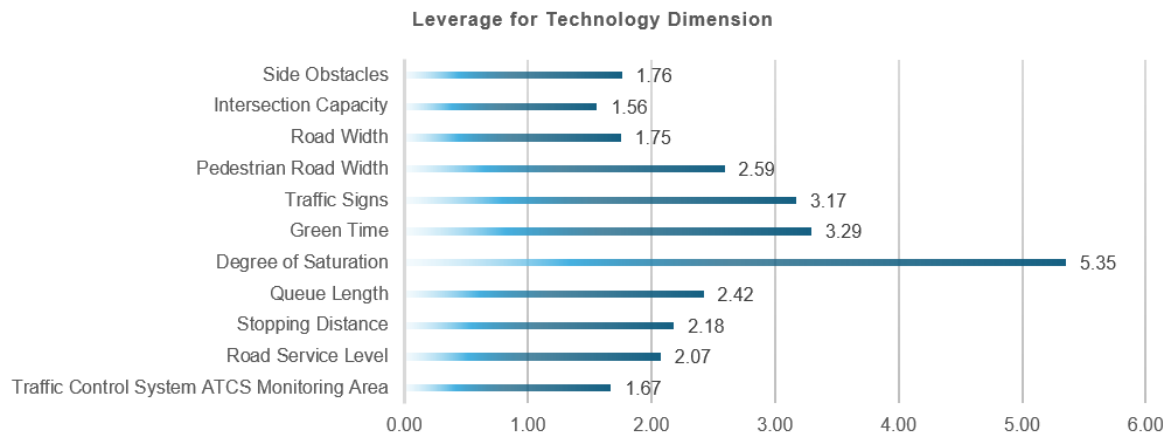


Figure 5. Leverage of the technology dimension

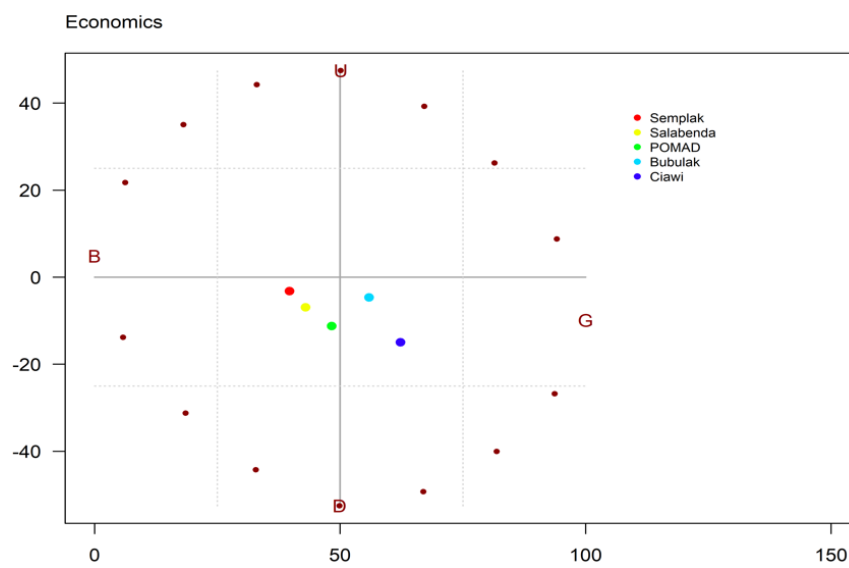


Figure 6. Economic sustainability ordination

3.3 Results and Discussion of the Ecological Dimension

The ecological dimension describes seven attributes: renewable fuel-based mass transportation, exhaust emission levels, noise levels, availability of green open spaces, availability of lanes for non-motorized vehicles, road width, and the availability of green lanes at each intersection. The results of the ecological dimension sustainability analysis, respectively, were as follows: the Salabenda signalized intersection (54.56%), the Semplak signalized intersection (51.79%), the Bubulak signalized intersection (57.06%), the POMAD signalized intersection (59.44%), and the Ciawi signalized intersection (43.71%). These results indicate that the POMAD signalized intersection performs best in the ecological dimension, with the highest sustainability score among the five intersections, as shown in Figure 8.

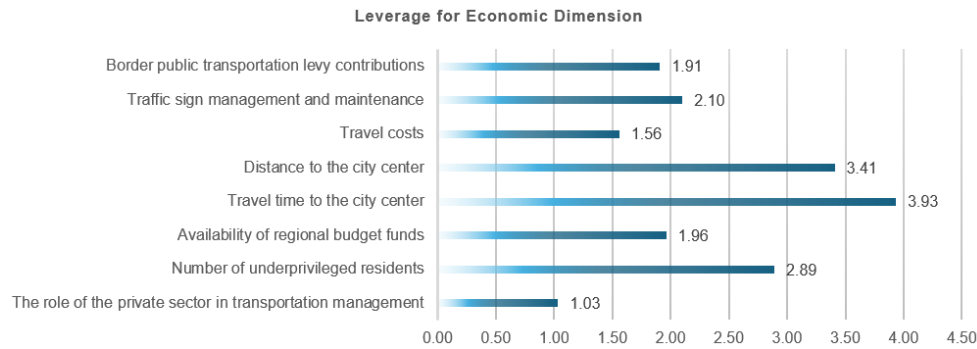


Figure 7. Leverage of the economic dimension

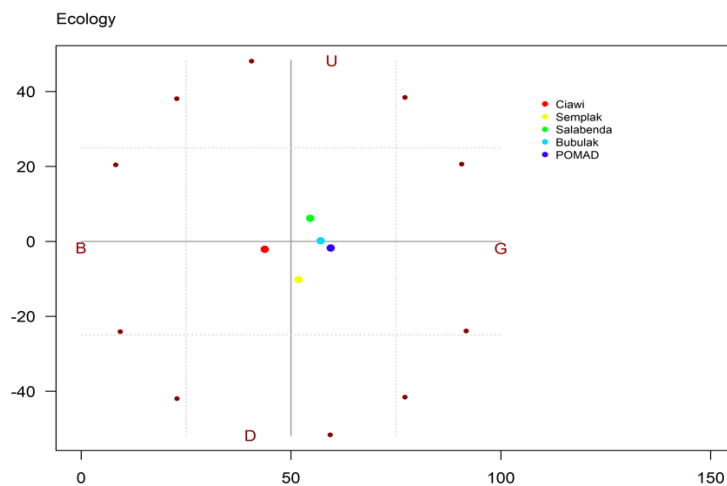


Figure 8. Ecological sustainability ordination

Furthermore, the ecological dimension of leverage is presented in Figure 9.

In the ecological dimension leverage model, the availability of lanes for non-motorized vehicles has a strong influence on the ecological dimension.

3.4 Results and Discussion of the Social Dimension

The social dimension is presented with nine attributes: level of public awareness/concern, number of accidents at intersections, pedestrian facilities for people with disabilities, use of hybrid vehicles, and public satisfaction with the level of service. The results of the sustainability analysis of the social dimension are, respectively, the Salabenda signalized intersection (57.19%), the Semplak signalized intersection (49.73%), the Bubulak signalized intersection (59.71%), the POMAD signalized intersection (55.59%), and the Ciawi signalized intersection (57.98%). These results indicate that the Bubulak signalized intersection performs most prominently in the social dimension, with the highest sustainability score among the five intersections, as shown in Figure 10.

Furthermore, the social dimension of leverage is presented in Figure 7. As shown in Figure 11.

The social dimension leverage model emphasizes public satisfaction with service levels, which has a strong influence on the social dimension.

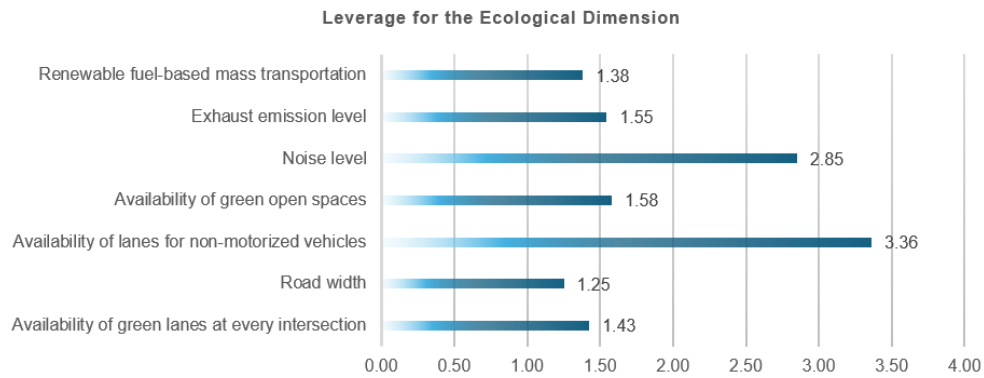


Figure 9. Leverage of the ecological dimension

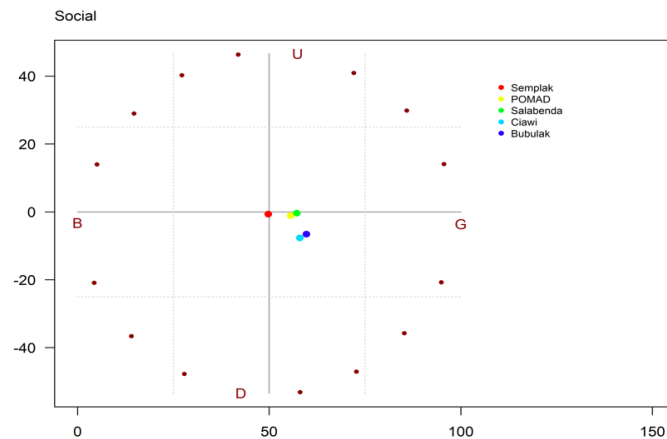


Figure 10. Social sustainability ordination

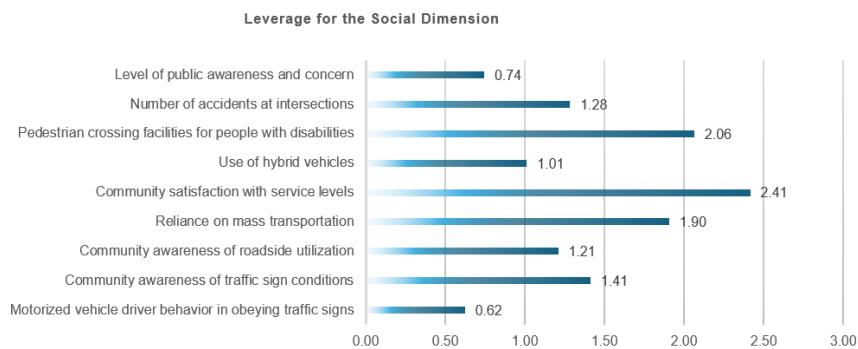


Figure 11. Leverage of the social dimension

3.5 Institutional Dimension Results and Discussion

The institutional dimension is presented with nine attributes: regional readiness in managing transportation infrastructure, border public transportation route permit management policies, legislative support for regional regulation development, cooperation agency formation scenarios, regulatory compliance, political support, government apparatus performance, e-government, and human resource capacity and capability. The results of the institutional dimension sustainability analysis are, respectively, at the Salabenda signalized intersection (48.11%), the Semplak signalized intersection (44.52%), the Bubulak signalized intersection (50.00%), the POMAD signalized intersection (35.87%), and the Ciawi signalized intersection (40.57%). Institutional sustainability at the Bubulak signalized intersection places a strong and prominent institutional aspect, as shown in Figure 12.

Furthermore, the institutional dimension of leverage is presented, as shown in Figure 13.

The institutional dimension leverage model emphasizes compliance with regulations and the improvement of border management policies.

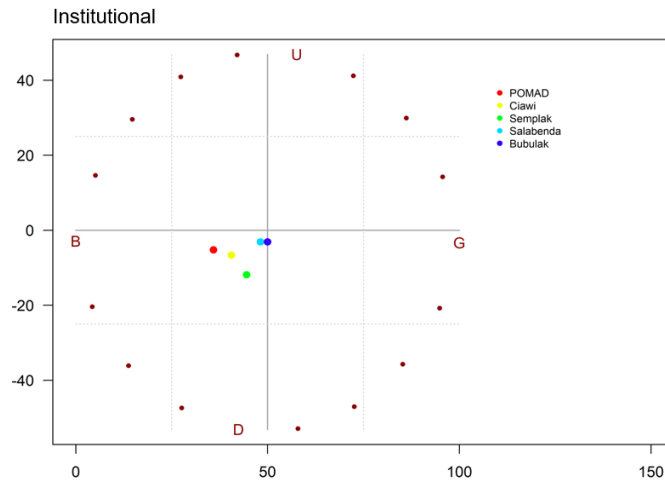


Figure 12. Institutional sustainability ordination

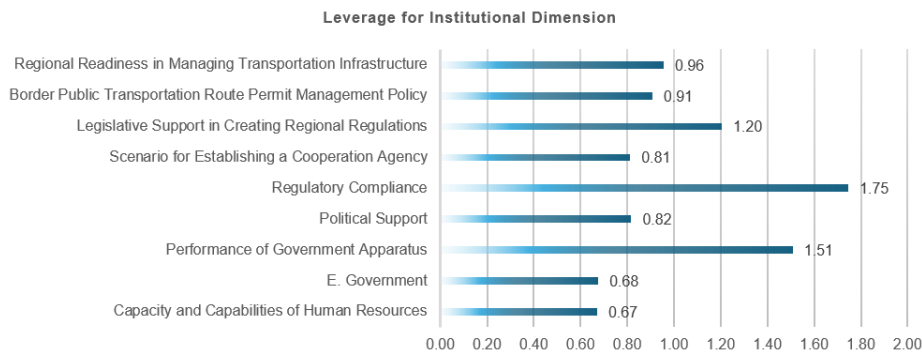


Figure 13. Leverage of the institutional dimension

3.6 Sustainability Radar Dimension

The level of sustainability of traffic engineering in the Bogor transportation system can be evaluated using a multidimensional analysis obtained from Rappfish. The results are shown in Figure 14.

Figure 14 shows that the technology dimension is the best aspect of sustainability compared to other dimensions. This is evident at the Salabenda intersection with the highest technology score of 72.33%, followed by Semplak with 62.36%. The social dimension is in the fairly good category, especially at the Bubulak intersection, with a score of 59.71%, indicating attention to community needs, such as accessibility for people with disabilities. In the ecological dimension, the POMAD intersection scored 59.44%, indicating that environmental aspects are beginning to be considered in intersection management. Meanwhile, the institutional dimension at Bubulak scored 50.00%, indicating that cooperation and coordination between agencies still need to be improved. The economic dimension has the lowest score, at 42.97% at the Ciawi intersection, so aspects of funding, cost efficiency, and economic benefits remain major challenges. Overall, these results indicate that technological development is quite good, but economic and institutional aspects still require improvement to support the sustainability of traffic-lighted intersections. Regional governments and inter-regional transportation agencies need to establish a border transportation coordination forum that regularly evaluates the performance of signalized intersections. Furthermore, a digital-based traffic data sharing system is needed to support synchronized signal settings and control vehicle queues across regions.

3.7 Discussion of Model Scenarios

Table 4 shows the optimistic scenario model pattern with interventions.

The diagram above shows four model scenarios through optimistic interventions on model parameters. The model parameters are used to realize a sustainable signalized intersection management model in border areas that has a strong impact on improving intersection performance. The concept is the SYSMODE. This SYSMODE scenario uses the optimistic scenario concept, namely the technological, social, ecological, institutional, and economic approaches with intervention patterns. The novelty of this research is the presence of SYSMODE. The higher economic value

of the Salabenda and POMAD intersections indicates that they have a smoother traffic flow compared to other intersections. This reduces vehicle delays and transportation operating costs, particularly for goods distribution and commuter travel. Furthermore, the locations of both intersections, located along key trade corridors and in urban access, support efficient economic mobility. The results of this study indicate that the technical dimension has a dominant influence on the sustainability of signalized intersections. This finding aligns with previous research that found signal timing and delay levels to be key indicators of urban intersection performance. However, this study differs by adding an institutional dimension to previously rarely analyzed border areas.

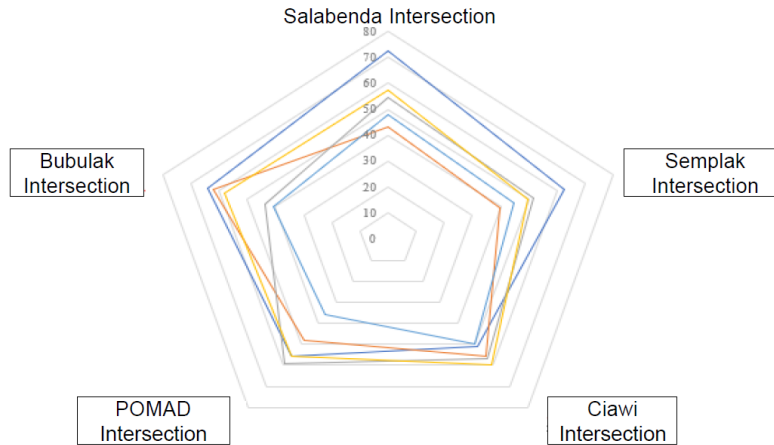


Figure 14. Sustainability radar diagram

Note: Figure 14 comparing the sustainability performance of five signalized intersections. The radial scale ranges from 0 to 80, with higher values indicating better sustainability performance. Blue line—higher performance at Salabenda Intersection and lower performance at POMAD and Ciawi intersections; orange line—relatively balanced performance across the intersections, with higher scores at Bubulak and Ciawi intersections; yellow line—relatively balanced performance across the intersections, with higher scores at Bubulak and Ciawi intersections.

Table 4. Recommendations with optimistic intervention model pattern scenarios

No.	Policy	Scenario Model	Intervention Pattern
1	Sustainability of intersection functions using a technological approach	1. Evaluation of green, yellow, and red phases, side obstacles, intersection gradients, and the presence of vehicles parked on the roadside. 2. Factors affecting intersection performance, regular signage management. 3. Signalized intersections offer numerous benefits for operators, vehicle users, and pedestrians.	Optimistic intervention
2	Cooperation in managing transportation infrastructure facilities	1. Optimizing transportation infrastructure development and border areas to meet interregional transportation needs. 2. Synchronizing transportation sector policies for border area development and economic development.	Optimistic intervention
3	Sustainability of accessibility control based on the characteristics of border areas	1. Implementing interregional transportation control to reduce accessibility disparities, in an integrated manner, and in accordance with regional characteristics. 2. Controlling and creating regulations regarding traffic engineering in the interregional transportation system.	Optimistic intervention
4	Policy in decision-making	1. Increasing cooperation in managing public facilities at intersections in border areas. 2. Increasing cooperation to minimize policy conflicts.	Optimistic intervention

4 Conclusions

The sustainability model for signalized intersections in border areas emphasizes the maintenance of existing technology while strengthening the social, ecological, institutional, and economic dimensions. Border intersections, particularly Ciawi and POMAD, require priority attention to improve transportation performance and sustainability. The proposed four-scenario model supports sustainable intersection management through signaling technology, integrated road and rail infrastructure management, accessibility control, and policy support. The novelty of this research lies in the development of the SYSMODE, which introduces a single-point system concept that supports all five sustainability dimensions. The implementation of the SYSMODE approach can improve the performance of signalized intersections for both road and rail transportation through integrated, measurable, and well-controlled management systems in border areas.

Author Contributions

Conceptualization, D.S.; methodology, D.S., S., and A.K.W.; validation, S. and A.K.W.; formal analysis, D.S., S., and A.K.W.; investigation, D.S.; data curation, D.S.; writing—original draft preparation, D.S.; writing—review and editing, S., A.K.W., and S.W.M.; visualization, D.S.; supervision, S. and S.W.M. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding authors upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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